

Alternative age estimation models for Canadian rockfish and salmon



Sclerochronology Laboratory

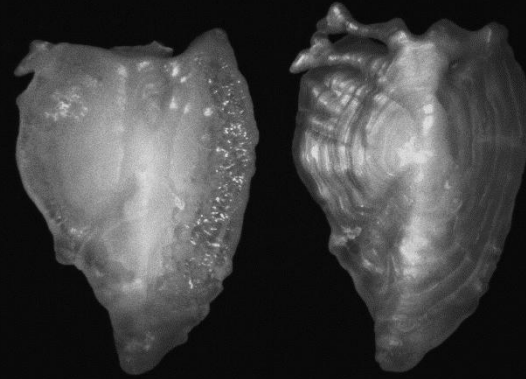
Alternative Age Estimation Research Branch

Fisheries & Oceans Canada
Pacific Biological Station

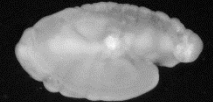
Sarah Gravel
Stephen Wischinowski

Ageing structures

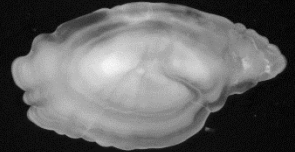
Otoliths – groundfish



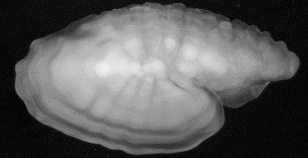
1 (1+)



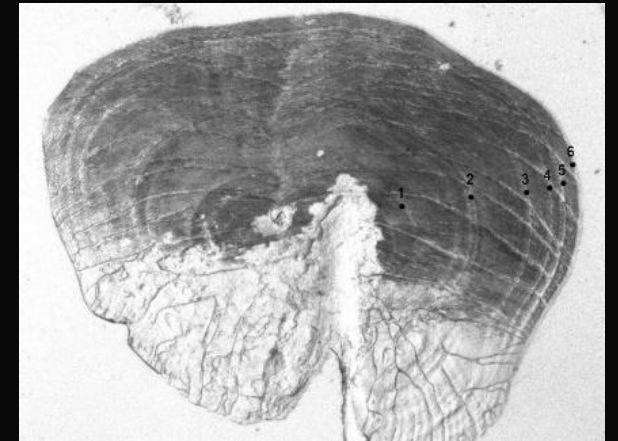
2 (2+)



3 (3+)



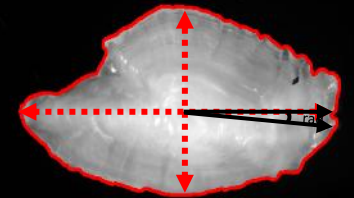
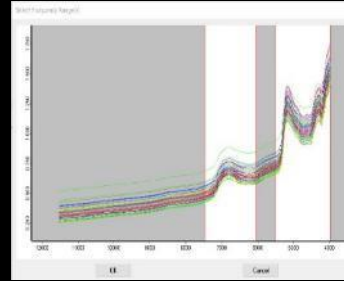
Scales – salmon and herring



Ageing structures

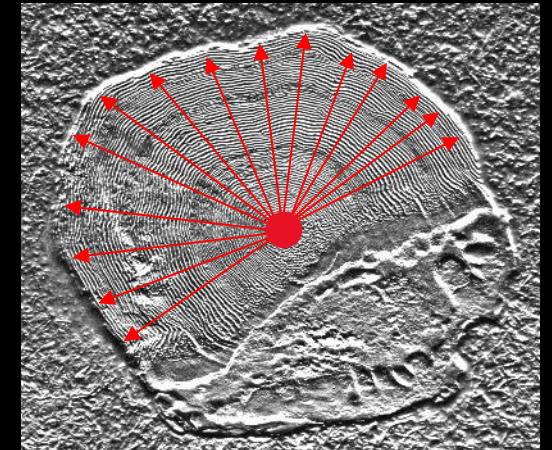
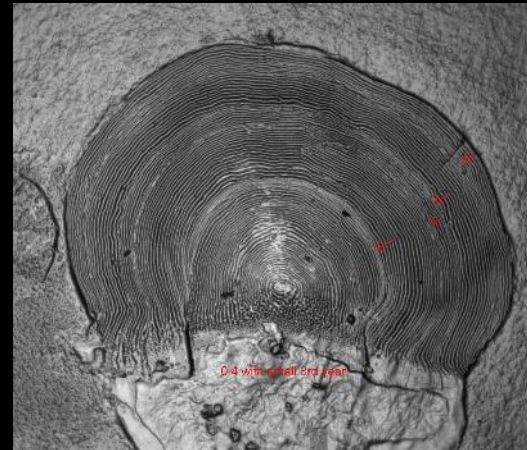
Otoliths – groundfish

- FT-NIR, weighing, imaging
- Spectra, weight, size, shape/otolith shape analysis (*ShapeR*)

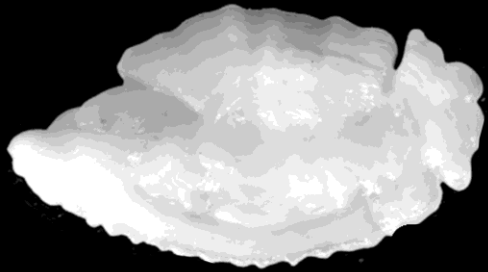


Scales – salmon and herring

- Imaging
- Growth patterns



Predicting rockfish ages from otolith morphometrics and NIR spectra (*Sebastes* spp.)



Pacific Ocean Perch
(*S. alutus*)

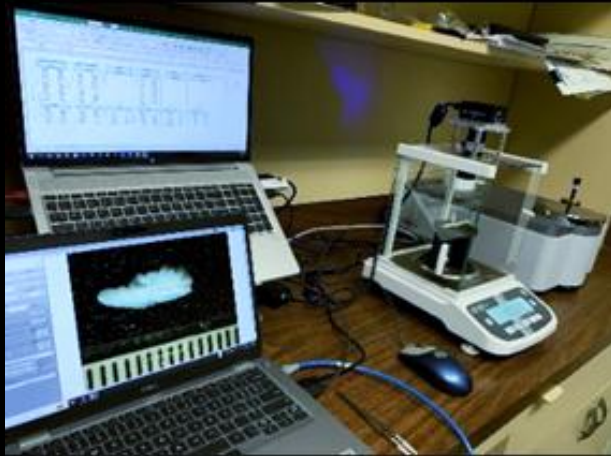


Rougheye & Blackspotted 'REBS' complex
(*S. aleutianus* & *S. melanostictus*)

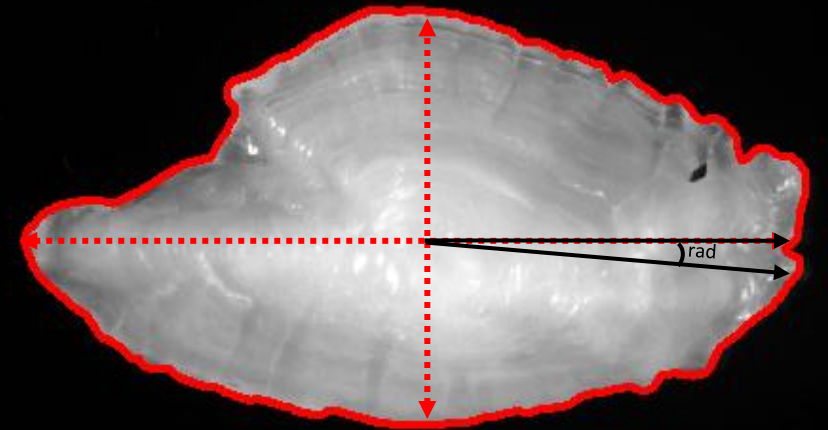


Yelloweye
(*S. ruberrimus*)

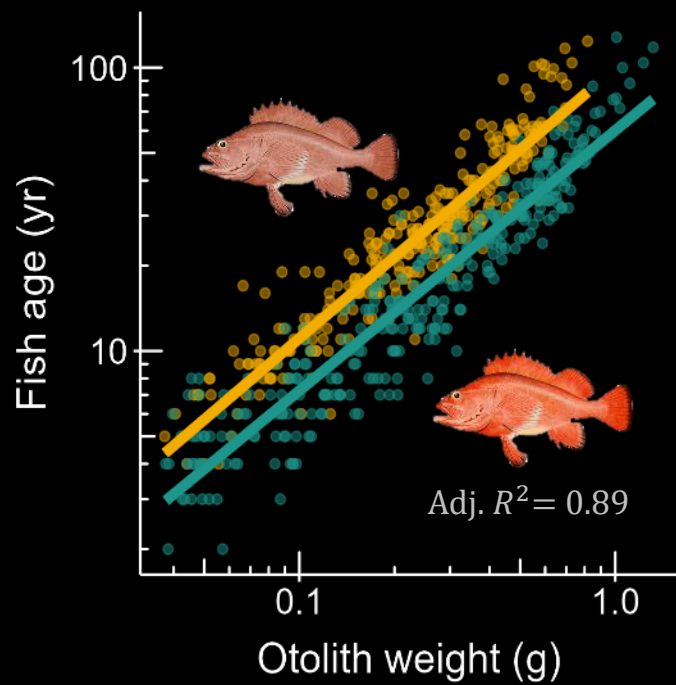
Otolith morphometrics: weight and size with *ShapeR*



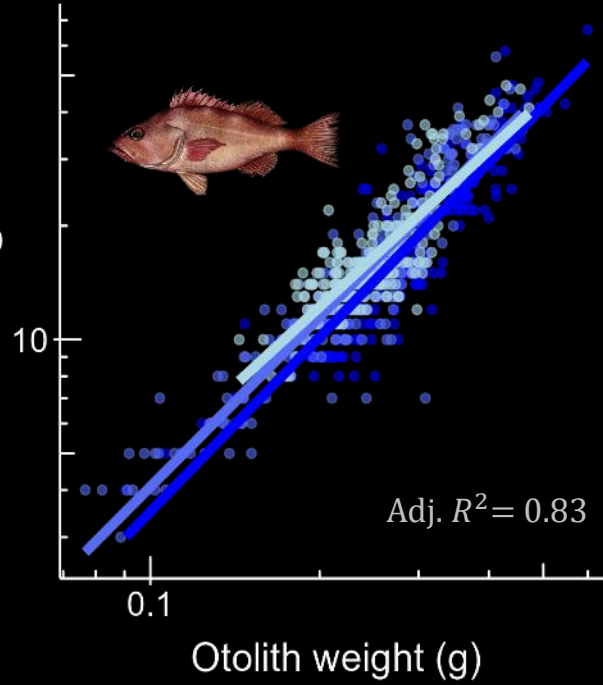
- Weight
- Length
- Width
- Area
- Perimeter



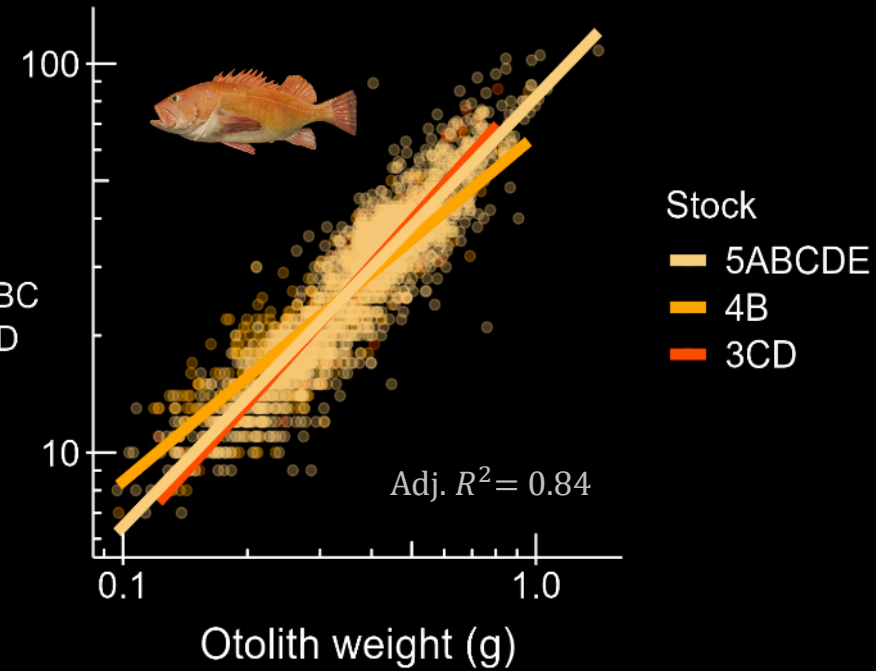
Rockfish age–otolith weight relationships



Rougheye & Blackspotted 'REBS' complex
(*S. aleutianus* & *S. melanostictus*)



Pacific Ocean Perch
(*S. alutus*)



Yelloweye
(*S. ruberrimus*)

Age ~ otolith weight + otolith size

Relative importance of predictors (Random Forest)



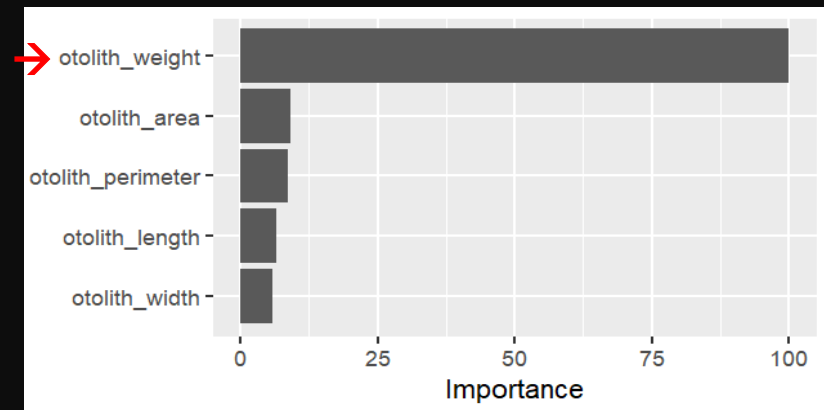
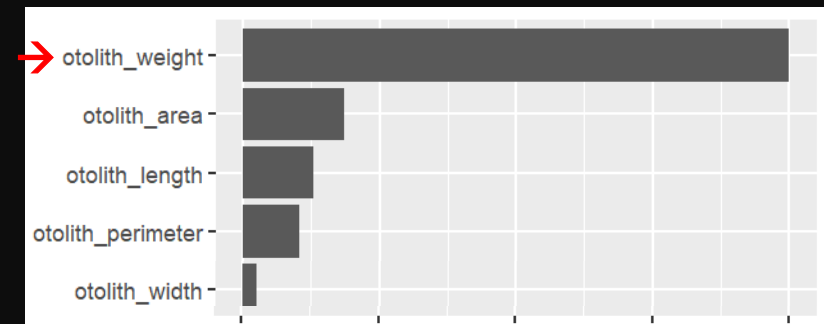
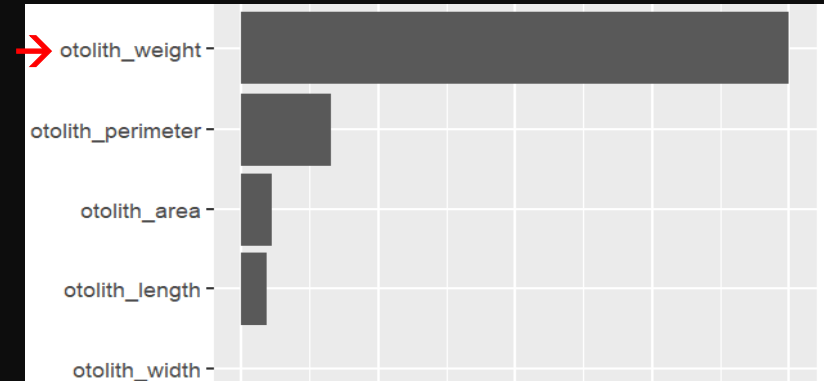
Rougheye
(*S. aleutianus*)



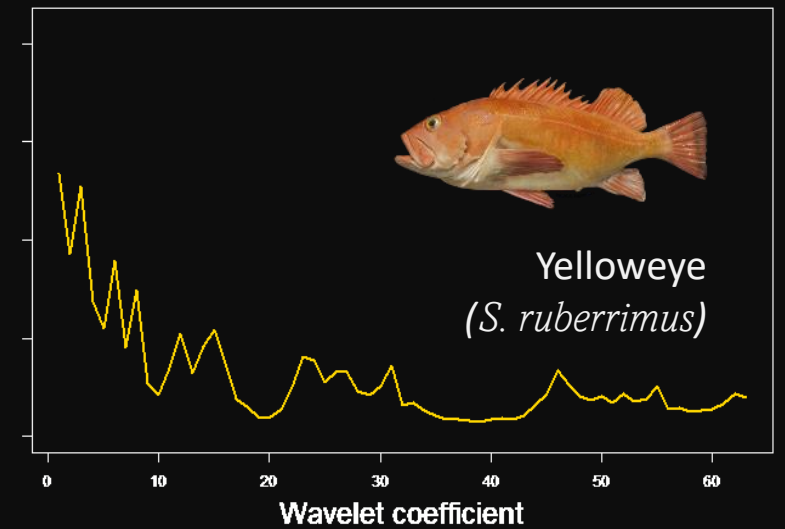
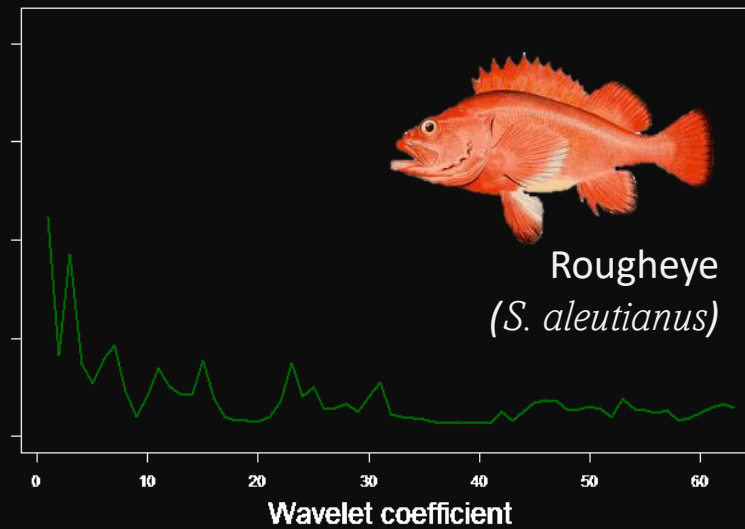
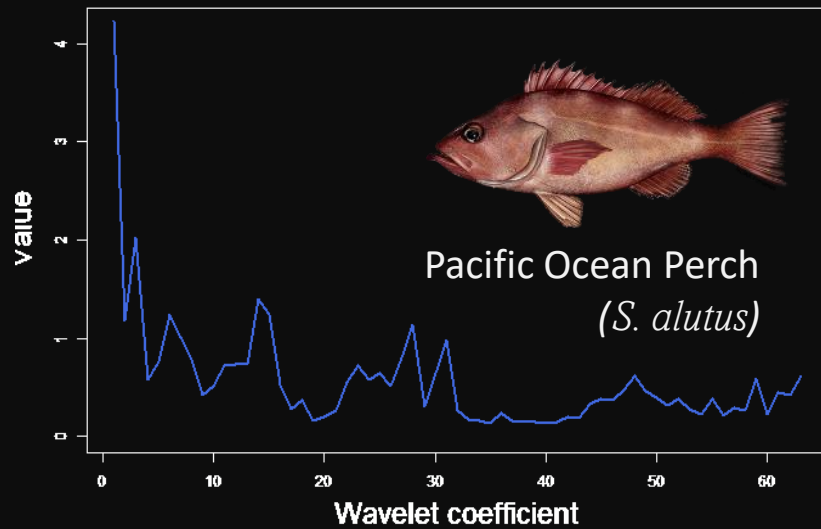
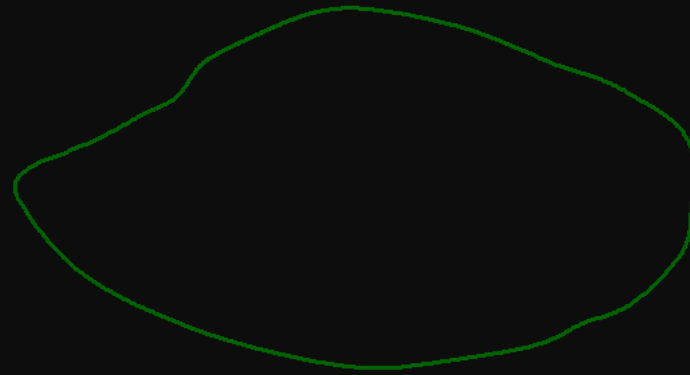
Pacific Ocean Perch
(*S. alutus*)



Yelloweye
(*S. ruberrimus*)



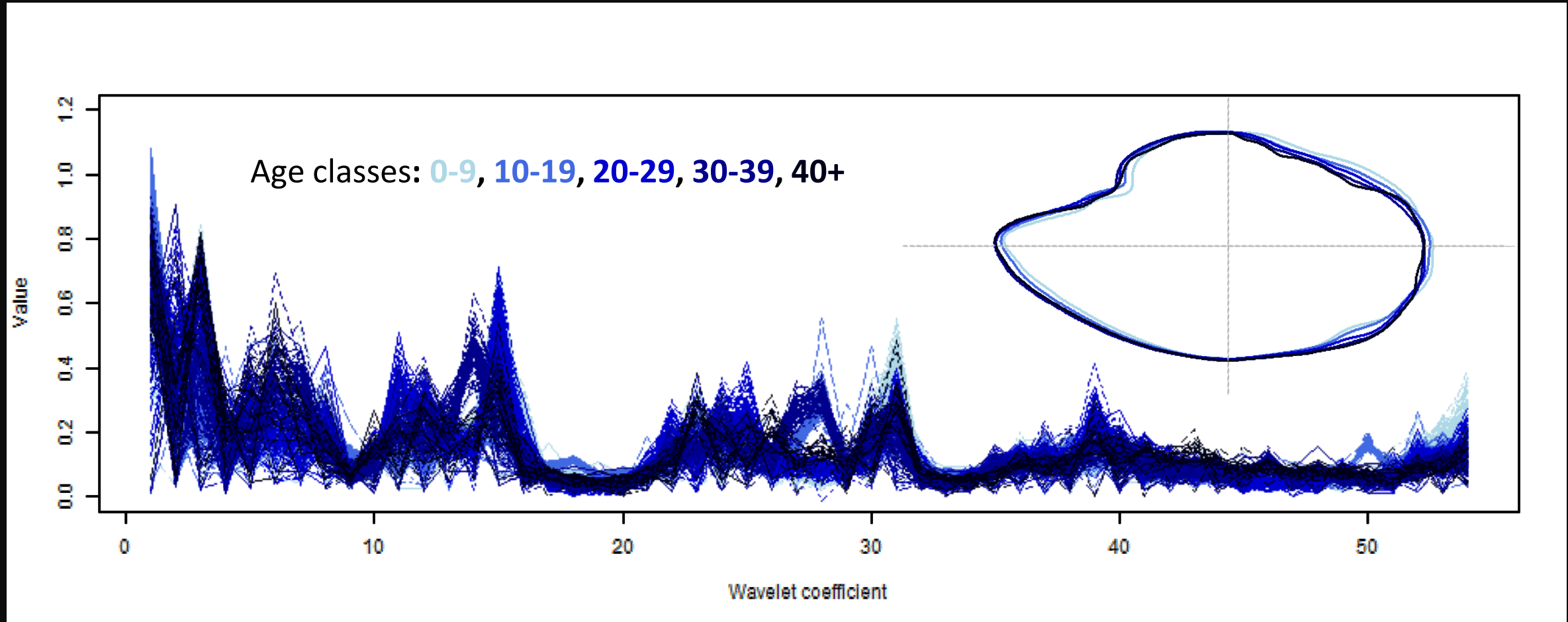
Otolith morphometrics: otolith shape analysis



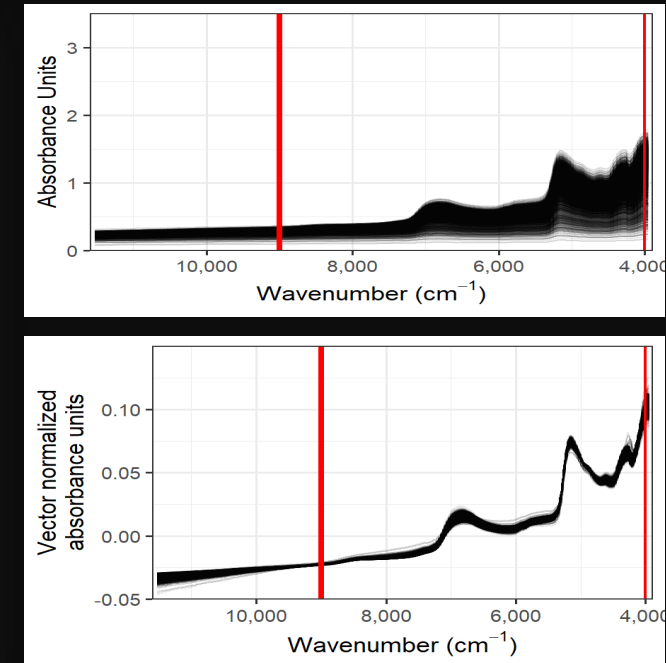
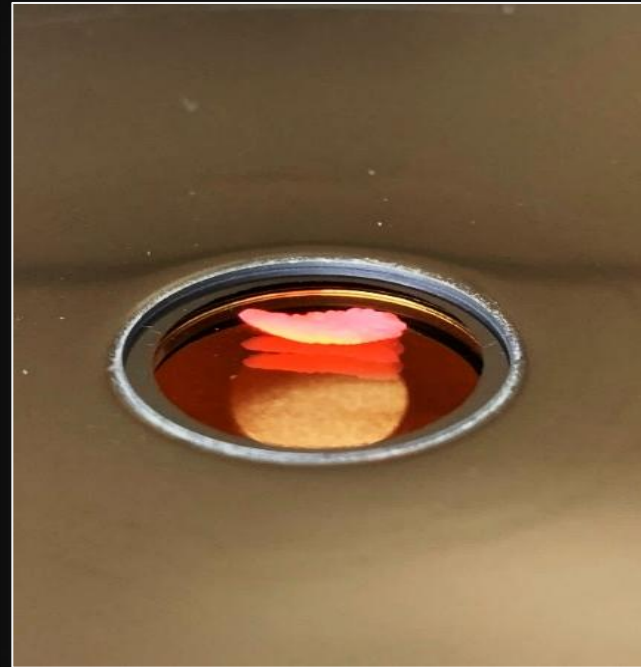
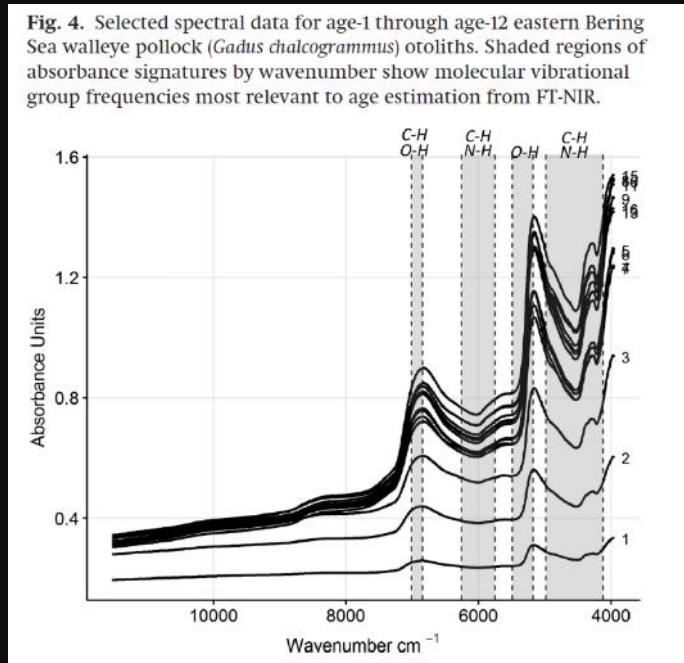
The relationship between age and otolith shape

Wavelet shape coefficients coloured by age class bins

Pacific Ocean Perch
(*S. alutus*)



Otolith composition: Fourier-transform near infrared spectroscopy

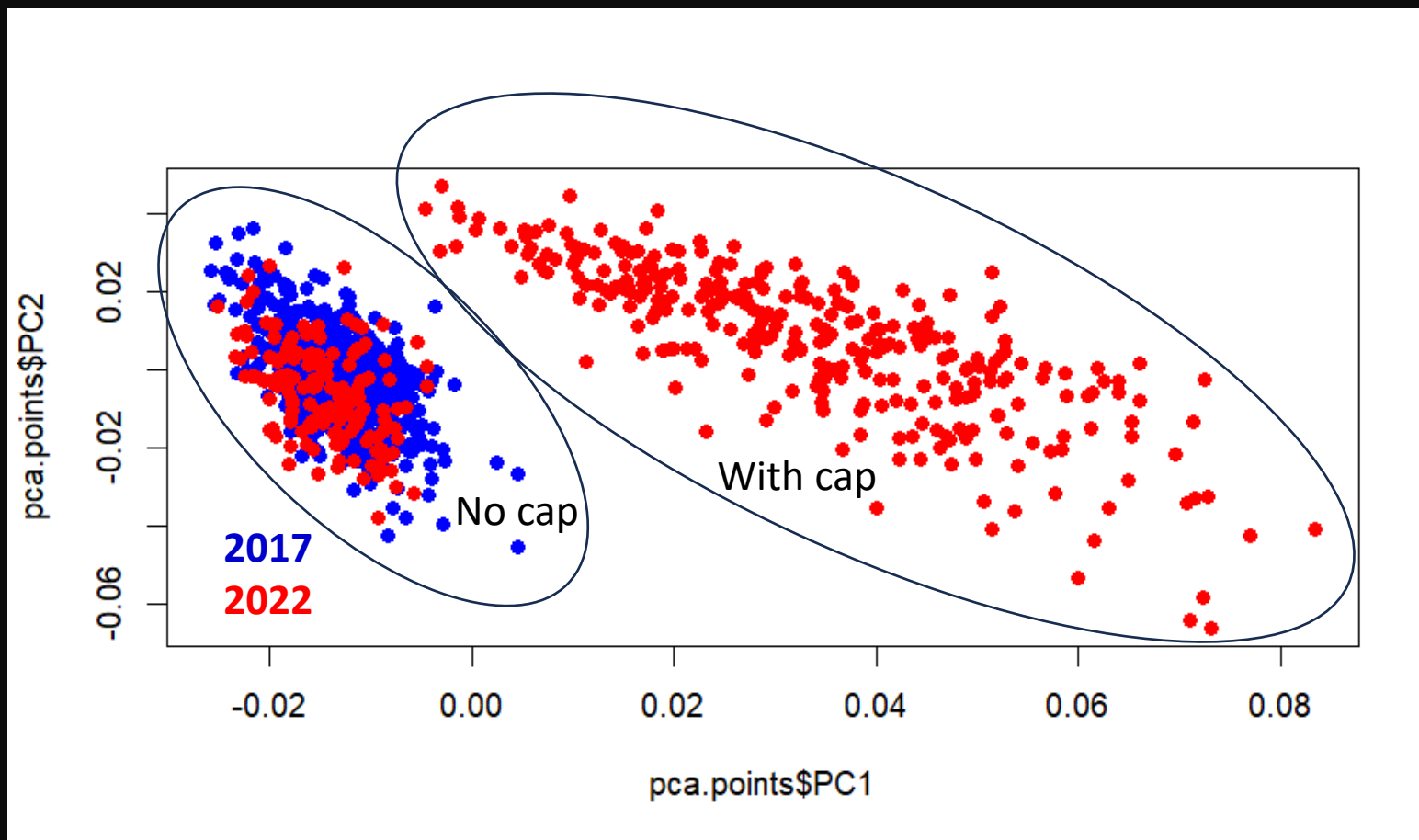


Helser et al. (2018)

FT-NIR PCA (methods matter)

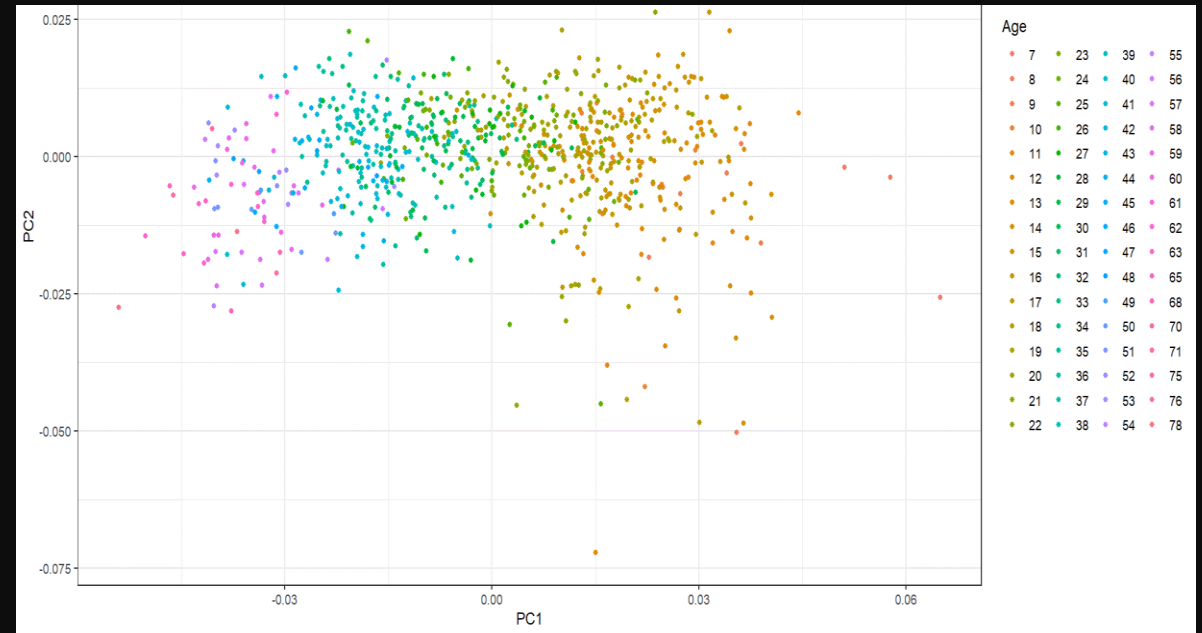
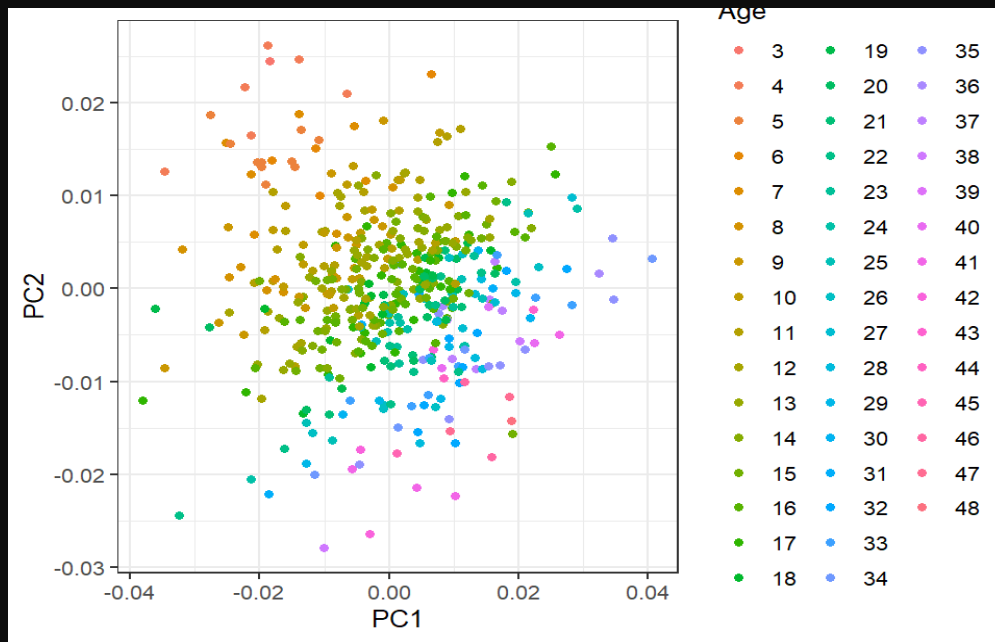


Pacific Ocean Perch
(*S. alutus*)



The relationship between age and otolith composition

FT-NIR PCA coloured by age

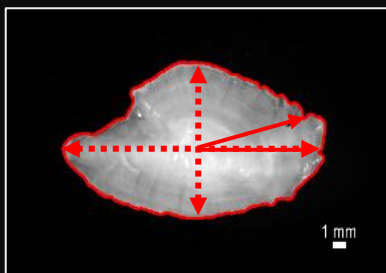


Pacific Ocean Perch
(*S. alutus*)



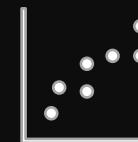
Yelloweye
(*S. ruberrimus*)





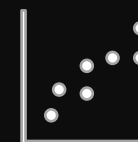
oto weight, size, shape

*Rougheye (/ Blackspotted)
'REBS' complex
age models*



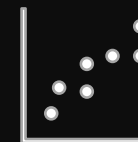
*oto weight, size, shape
oto FT-NIRs*

*Pacific Ocean Perch
age models*



*oto weight, size, shape
oto FT-NIRs*

*Yelloweye
age models*

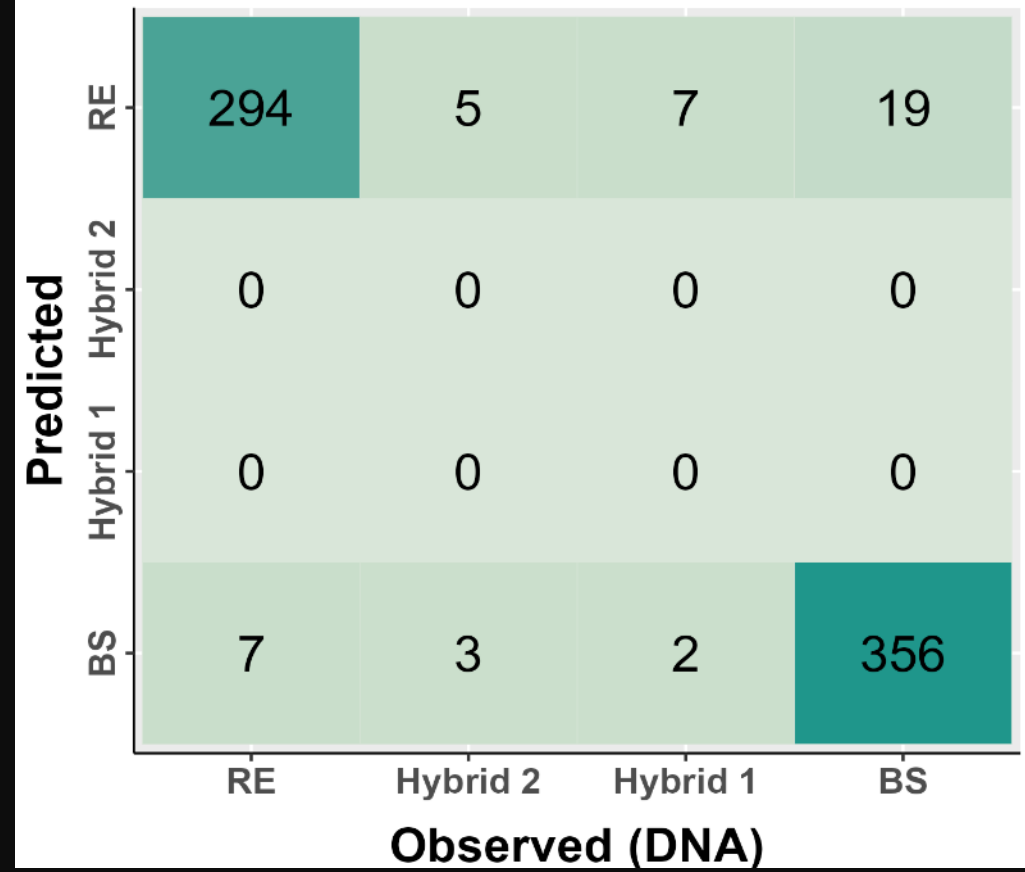


REBS species classification



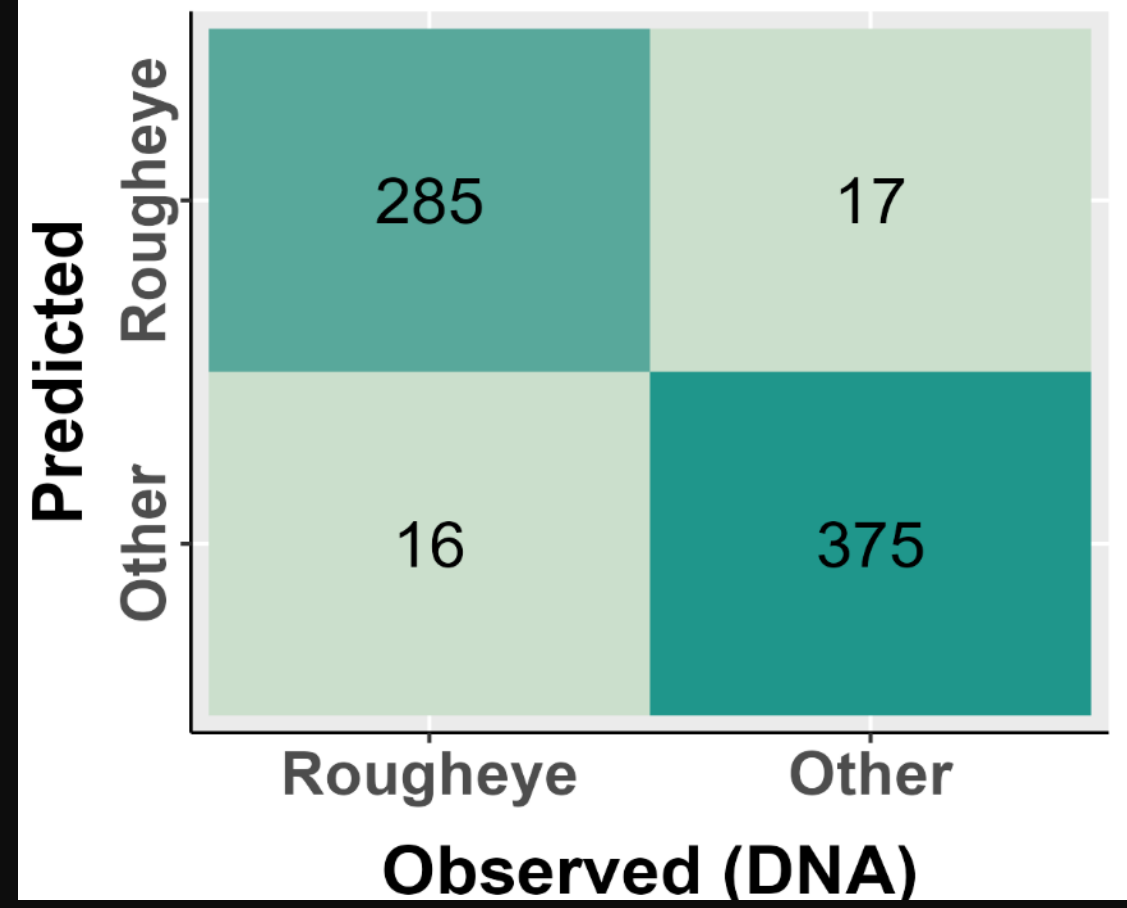
Which REBS species is this?
(Rougheye, Blackspotted, or a hybrid)

*** Accuracy for RE 0.95 ; for BS 0.96**
*** Precision for RE 0.9 ; for BS 0.97**



Is this fish a Rougheye?

*** Accuracy 0.95 * Precision 0.94**



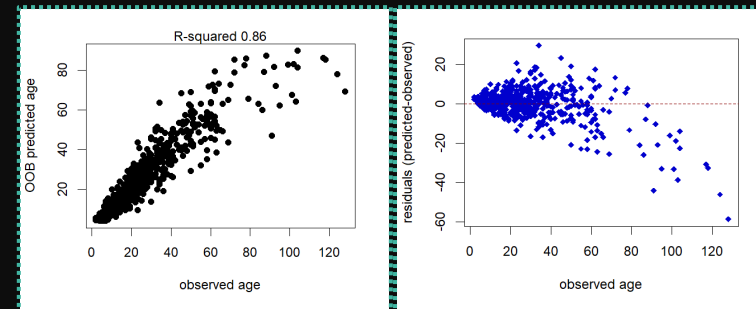
Fish age ~ otolith weight + otolith size + otolith shape + NIR spectra*

(*except REBS: no FT-NIR data for our aged REBS samples)



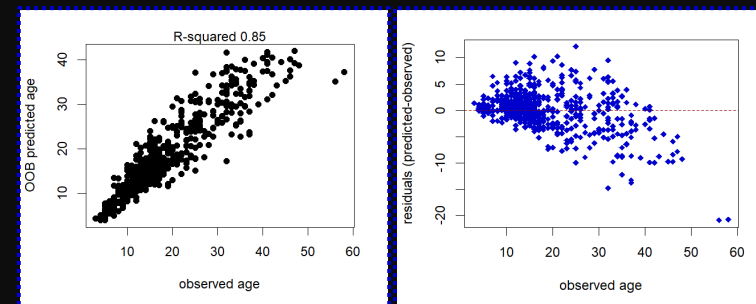
Rougheye

Adj. $R^2 = 0.86$
76% of ages ± 5 yrs



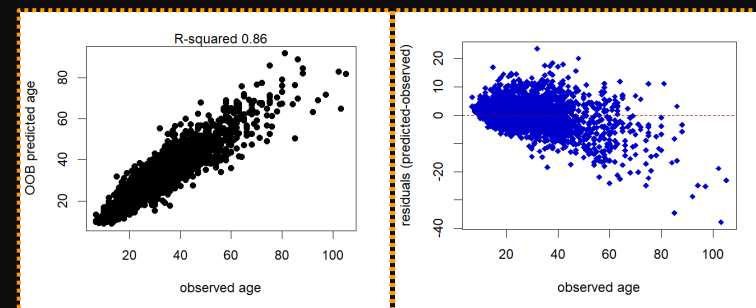
Pacific Ocean Perch

Adj. $R^2 = 0.85$
78% of ages ± 1 yr



Yelloweye

Adj. $R^2 = 0.86$
71% of ages ± 2 yrs



Precision Targets for Species

Agreement Target	% +/-0	% +/-1 yr	% +/-2 yrs	% +/-3 yrs	% +/-5 yrs
Rockfish					
<i>S. alascanus</i>					80
<i>S. aleutianus</i> (Rougheye)					60-80
<i>S. altivelas</i>					80
<i>S. alutus</i> (Pacific Ocean Perch)		80			
<i>S. babcocki</i>			70		
<i>S. brevispinis</i>		50	70		
<i>S. caurinus</i>		80			
<i>S. entomelas</i>		80			
<i>S. flavidus</i>		80			
<i>S. maliger</i>		70			
<i>S. pinniger</i>		50	80		
<i>S. proriger</i>		80			
<i>S. reedi</i>			80		
<i>S. ruberrimus</i> (Yelloweye)			70-80		
<i>S. wilsoni</i>		90			



76 % $\pm$ 5 yrs

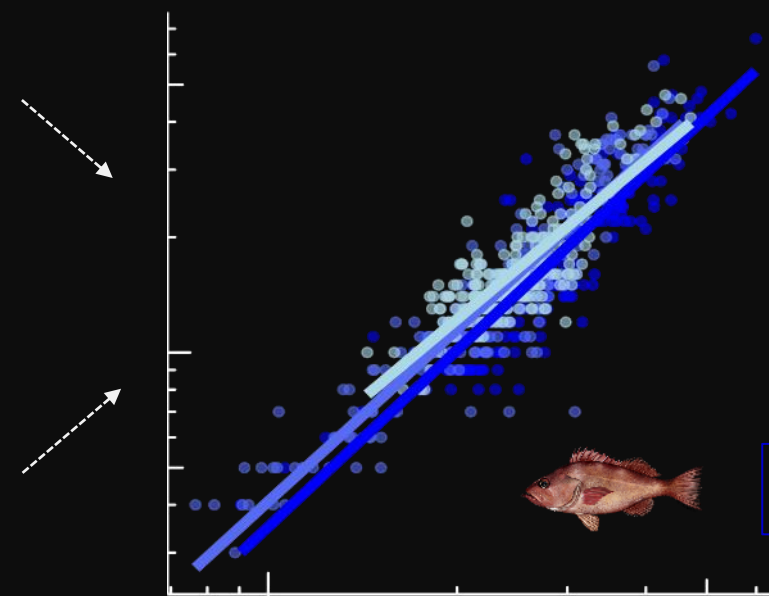
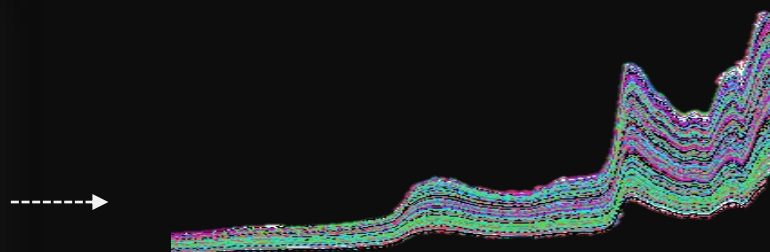
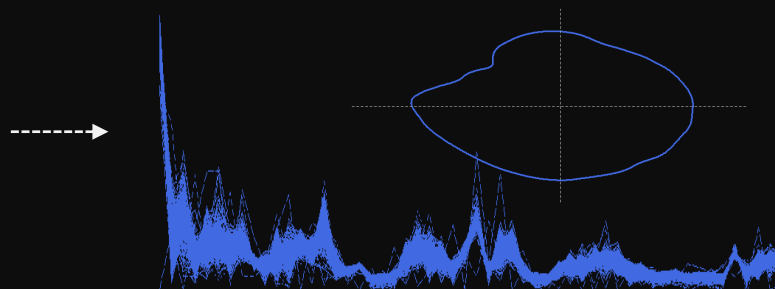
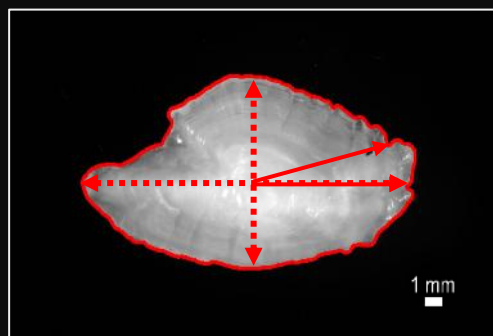


78 % $\pm$ 1 yr



71 % $\pm$ 2 yrs

package : *fishoto*



$78 \% \pm 1 \text{ yr}$

Predicting Chum salmon ages from scales



Chum salmon
(*Oncorhynchus keta*)

Employing deep learning for chum salmon (*Oncorhynchus keta*) age estimation through scale image analysis

(Manuscript submitted to CJFAS January 2026)

A computer vision model (“Chumper Vision”) and workflow for the automated prediction of the ages of Chum salmon scales from high resolution images ;

A total of 66,137 **previously-aged** scale images were collected from four agencies: NOAA, ADF&G, NSRAA, and DFO

- 92% were age 4 and 5 (large age class imbalance)
- Subsampling to minimize bias towards dominant age classes
- 15,754 images (24%) were split into three datasets for training (n = 11,012; 70%), validation (n = 2,365; 15%), and testing (n = 2,377; 15%) ;

Three CNN models (EfficientNet v2L, NASNetLarge, and ConvNeXt) were trained independently, using images pre-processed by two models: *You Only Look Once* (YOLO) or *Segment Anything Model* (SAM)

AI server (“Skynet”) running Linux Ubuntu 22.04 and dedicated Anaconda environments with Python (3.10.18), TensorFlow (2.13.0), and Keras (2.13.1)

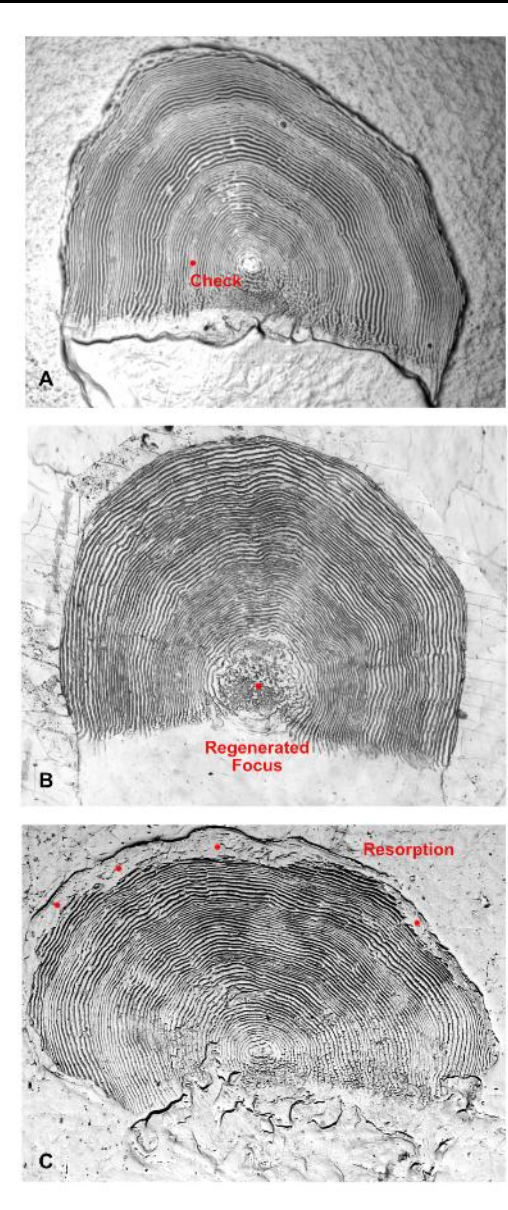
Scale images

Selected only high-quality images, scales with minimal resorption

Image prep and standardization (image segmentation models, converted to three-channel RGB matrices, resized to 512 x 512, normalized)

Signal balancing

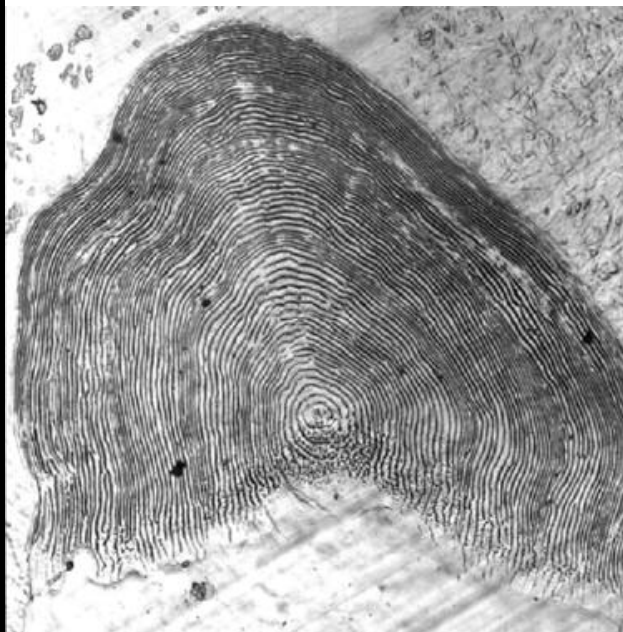
Laplacian function was used to exclude poor quality images (<300)



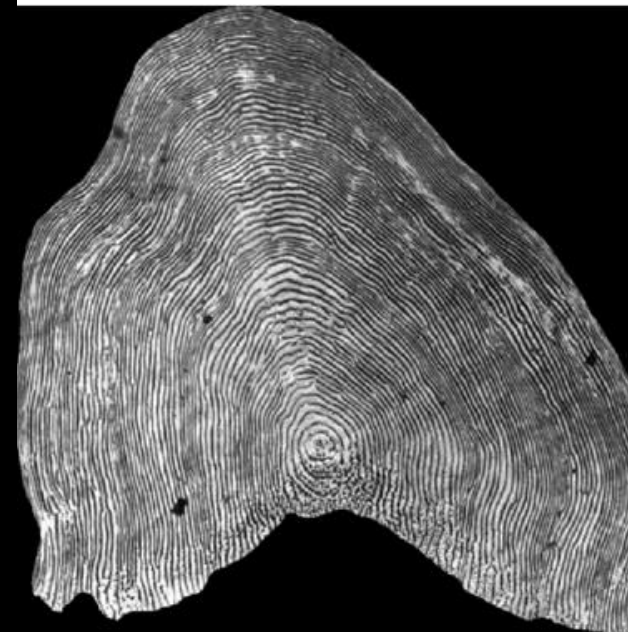
Raw
3352×4425



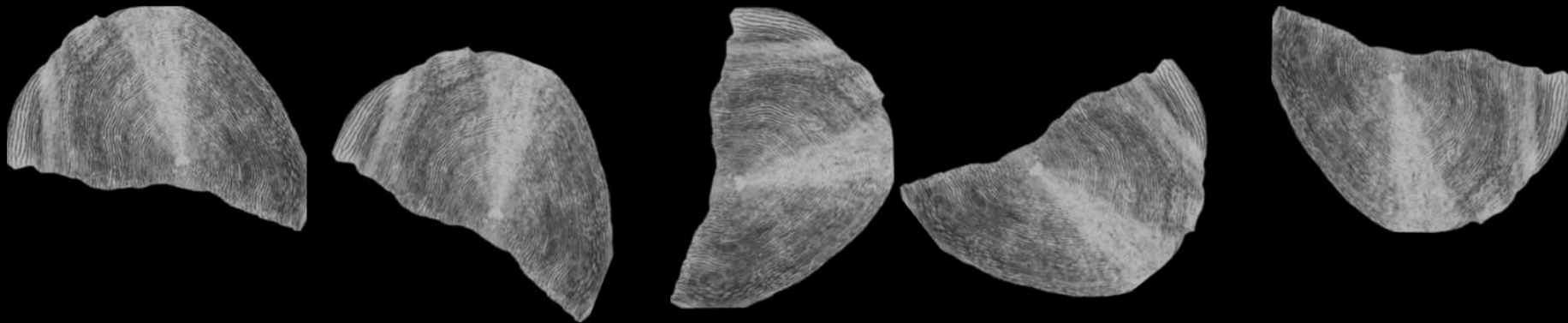
YOLO Cropped (Resized)
512×512



SAM (Resized)
512×512



Augmentation of scale images provided the models with more training data by creating multiple distorted versions of each image to prevent overfitting.



Convolutional Neural Network Model

Input: High-resolution images of previously aged scales

Convolutional Layer:

- Application of filters to the scale, sliding them over the image to detect patterns.
- Each filter is designed to recognize specific features, like the spacing of circuli, annuli and checks.
- This process creates a hierarchical representation: early layers detect simple features, while deeper layers combine them to identify more complex patterns.

Pooling Layer:

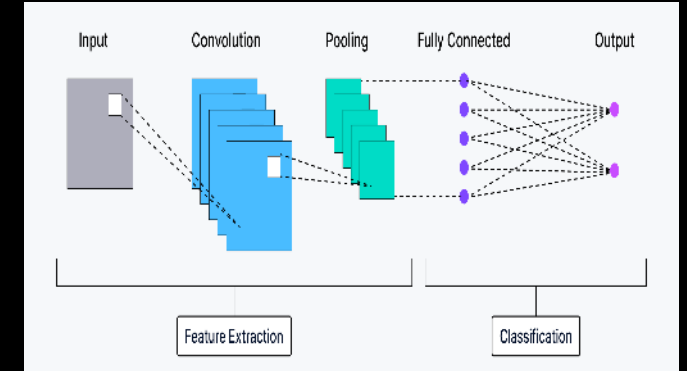
- This layer reduces the spatial dimensions making the model more robust and less sensitive to the exact location of features in the image

Fully Connected Layer:

- After several convolutional and pooling layers, the extracted features are "flattened" into a one-dimensional vector.
- This vector is then fed into a fully connected layer, where every neuron is connected to every neuron in the next layer.

Output Layer:

- The final layer provides the predicted age, often a probability distribution across different age classes



An Ensemble model was used to combined all three models. The final ensemble predictions, the weighted average of the three models, were rounded to the nearest integer and compared to the age readers' age estimates.

Age Class 1 - Samples: 31, Precision: 100.00%

Age Class 2 - Samples: 16, Precision: 44.00%

Age Class 3 - Samples: 555, Precision: 94.31%

Age Class 4 - Samples: 721, Precision: 94.09%

Age Class 5 - Samples: 492, Precision: 91.17%

Age Class 6 - Samples: 12, Precision: 60.00%

Age Class 7 - Samples: 2, Precision: 0.00%

Combining age 3 to 6-year-old samples, accuracy across agencies was high: NOAA (89.2%), ADFG (96.2%), NSRAA (85.0%), and DFO (94.1%).

Model performance: MSE

YOLO vs. SAM pre-processing

Images from unpublished manuscript (in review) have been removed

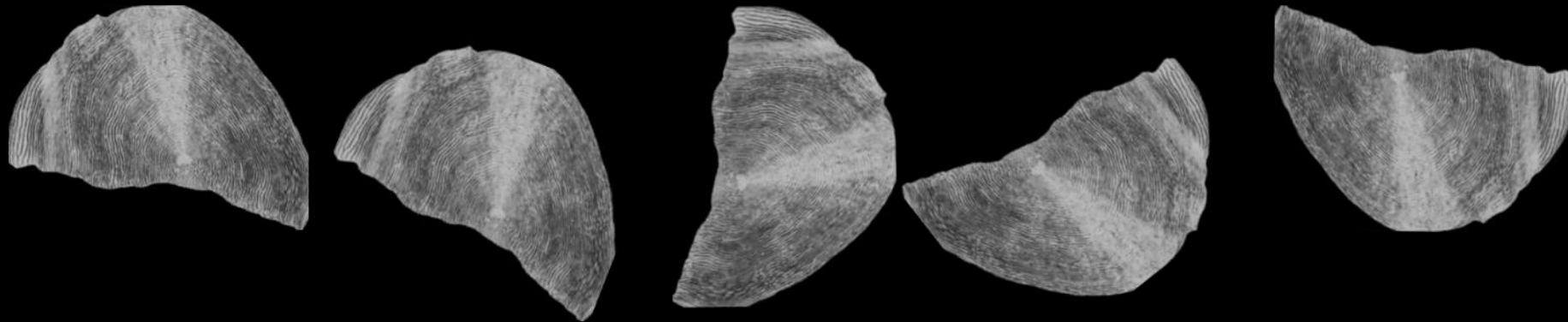
Model performances: prediction accuracy

EfficientNetV2L, ConvNeXtLarge, NASNetLarge, and the Ensemble model

Images from unpublished
manuscript (in review) have
been removed

Where are we going with this work?

1. Fine-tuning the current model (training on an additional 8000+ BC Chum salmon)
2. Prerequisites to operationalization case study/proof of concept (establishing *Framework* laptop workflow)

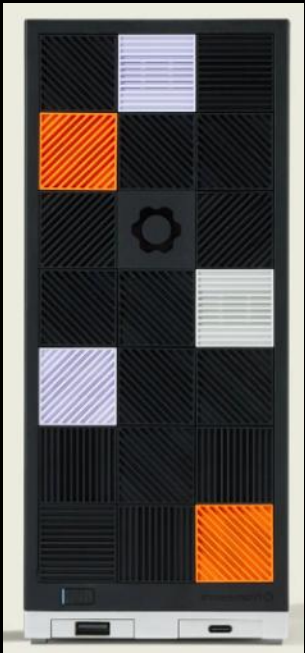


Domain shift & the importance of fine tuning

Model accuracy of the
Original EfficientNet Model
vs.
Fine-tuned EfficientNet Model
with additional training on
Alaska DIPAC images

age_class	support	orig_acc	finetuned_acc	delta
1	0	NaN	NaN	NaN
2	0	NaN	NaN	NaN
3	44	0.795455	0.931818	0.136364
4	438	0.874429	0.949772	0.075342
5	205	0.712195	0.853659	0.141463
6	2	0.000000	0.000000	0.000000
7	0	NaN	NaN	NaN

Framework computer case study: Automated ageing workflow for Chum

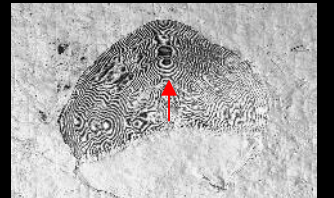


Launch Leica high speed imaging software
Image all scales in the scale card*
*(*currently selecting just one scale per fish)*

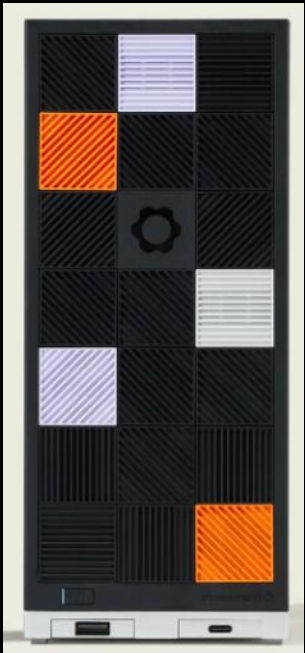
Measurement first year marine growth

Scale images & data through processing pipelines

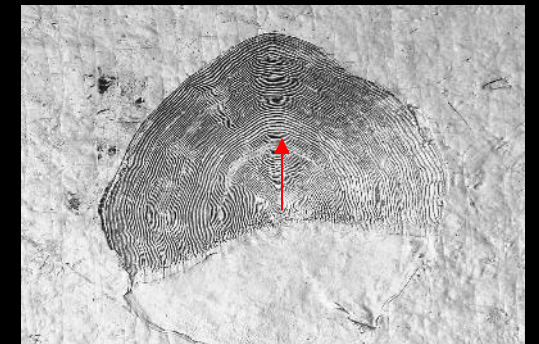
Predict scale ages, using pre-loaded model weights*
*(*model training on Linux server "Skynet" or Dell Precision 7960 supercomputer)*



Next steps



- Training/testing on highly resorbed scales (often seen in terminal fisheries)
- Incorporating additional variables (sacrifice date, location)
- Chinook (+ other salmon species such as Sockeye, herring)
- Modeling 1st year marine growth
- Integration & operationalizing Chum workflows
- Saliency mapping





PACIFIC SALMON
STRATEGY INITIATIVE
(PSSI)



**Sclerochronology
Laboratory**

Fisheries & Oceans Canada
Pacific Biological Station

Canada



Fisheries and Oceans Canada
Pêches et Océans Canada